H. Kaan Kale

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Research Interests: Mechanistic Interpretability, State Space & Hybrid Language Models, Efficient Large-Scale Training, Privacy-Preserving Representations

Available for internships: Summer 2027

Bio

Ph.D. student in Machine Learning at Georgia Tech, advised by Prof. Sriram Vishwanath. I study how language models store and retrieve information. Publications at ICLR, ISIT, Interspeech, CoRL, and GLOBECOM.

Education

Georgia Institute of Technology — Jan 2026 – Present
Ph.D. in Machine Learning, School of Electrical and Computer Engineering — Atlanta, GA, USA
GPA: 4.0/4.0

Bogazici University — 2020 – 2025
B.Sc. in Electrical & Electronics Engineering, Specialization: Signal Processing — Istanbul, Turkey

Research & Work Experience

Protopia AI — May 2026 – Aug 2026
Research Engineer Intern, mentored by Kyle Mylonakis & Eiman Ebrahimi — London, UK

- Co-developed distributed training and fine-tuning infrastructure for **NVIDIA Nemotron-3 Super and Ultra** on multi-node GPU clusters (FSDP, expert parallelism), adopted by the team for its training runs.
- Enabled **6**× longer-context training through memory optimization; discovered a bug in **mamba-ssm** that silently corrupted model state beyond **∼60k tokens**, leading to an upstream fix (under review).
- Integrated **4**× embedding compression into the company's privacy-preserving LLM product, now in production, reducing client–server traffic at the best size–accuracy trade-off.

Georgia Institute of Technology — Aug 2025 – Present
Graduate Researcher, advised by Prof. Sriram Vishwanath — Atlanta, GA, USA

- Led an interpretability study of how Transformer and hybrid (recurrent + attention) language models retrieve facts in multi-step questions, across **12 models** (first author, under review at **ICLR 2027**).
- Showed that prompt-injection probes mostly learn conversation structure, not the injection; matched negative training data cut false alarms across six open models (first author, submitted to **NAACL 2027**).
- Developed the joint embedding model for **ProofBridge**, a framework that translates natural-language math proofs into formal, machine-checkable proofs, improving retrieval accuracy by **3.28**× (**ICLR 2026**).

Protopia AI — Nov 2024 – Dec 2025
Research Contractor (Remote), mentored by Kyle Mylonakis & Eiman Ebrahimi — Austin, TX, USA

- Researched privacy-preserving LLM representations, evaluating the utility–privacy trade-off of learned embedding obfuscation (**Stained Glass Transform**, arXiv 2025).
- Developed **BeamClean**, a language-model-guided reconstruction attack, showing that the company's product is more robust than traditional obfuscation methods (first author, arXiv 2025).
- Built scalable PyTorch evaluation infrastructure for reconstruction and model-inversion experiments.

The University of Texas at Austin, LINC Lab — Aug 2023 – Dec 2024
Undergraduate Researcher, advised by Profs. Sriram Vishwanath & Muriel Médard (MIT) — Austin, TX, USA

- Researched privacy and compression of language-model text representations.
- Led **TexShape** as first author: information-theoretic sentence embeddings that hide sensitive attributes while preserving task utility; designed the framework and ran all experiments (**ISIT 2024**).
- Contributed to **MultiTok**, LZW-inspired variable-length tokenization for efficient LLMs, and **WavShape**, information-theoretic privacy for speech (Interspeech 2025).

The University of Texas at Austin, Swarm Lab — Jun 2023 – Dec 2024
Undergraduate Researcher, advised by Prof. Sandeep Chinchali — Austin, TX, USA

- Researched resource allocation, data collection, and robustness for fleets of learning agents.
- Modeled and implemented early-exit neural networks for image and text, providing the accuracy–compute analysis behind a fair resource-allocation framework (**GLOBECOM 2025**).
- Co-designed the **SecureSpectra** deepfake-defense pipeline, validating its high-frequency signature hypothesis across multiple deepfake models and voice datasets (**Interspeech 2024**).

Conference & Workshop Papers

- Jana, P., **Kale, K.**, Tanriverdi, A. E., Song, C., Vishwanath, S., & Ganesh, V. **ProofBridge: Auto-Formalization of Natural Language Proofs in Lean via Joint Embeddings.** In *International Conference on Learning Representations (ICLR)*, 2026.
- Baser, O., **Kale, K.**, Li, P.-H., & Chinchali, S. **Fair Resource Allocation for Fleet Intelligence.** In *IEEE Global Communications Conference (GLOBECOM)*, 2025.
- Baser, O., Tanriverdi, A. E., **Kale, K.**, Chinchali, S., & Vishwanath, S. **WavShape: Information-Theoretic Speech Representation Learning for Fair and Privacy-Aware Audio Processing.** In *Proceedings of Interspeech*, 2025.
- **Kale, K.**, Esfahanizadeh, H., Elias, N., Baser, O., Médard, M., & Vishwanath, S. **TexShape: Information Theoretic Sentence Embedding for Language Models.** In *IEEE International Symposium on Information Theory (ISIT)*, 2024.
- Baser, O., **Kale, K.**, & Chinchali, S. **SecureSpectra: Safeguarding Digital Identity from Deep Fake Threats via Intelligent Signatures.** In *Proceedings of Interspeech*, 2024.
- Akcin, O., Tanriverdi, A. E., **Kale, K.**, & Chinchali, S. **Fleet Supervisor Allocation: A Submodular Maximization Approach.** In *Conference on Robot Learning (CoRL)*, 2024.
- Elias, N., Esfahanizadeh, H., **Kale, K.**, Vishwanath, S., & Médard, M. **MultiTok: Variable-Length Tokenization for Efficient LLMs Adapted from LZW Compression.** Poster, *Learn to Compress & Compress to Learn Workshop, ISIT*, 2026.

Preprints & Under Review

- **Kale, K.** & Vishwanath, S. **Prompt-Injection Probes Learn the Conversation, Not Just the Injection.** Under review at *ARR* (targeting *NAACL 2027*). arXiv:2610.18408.
- **Kale, K.**, Baser, O., & Vishwanath, S. **The Key Handoff: Retrieval in Hybrid Language Models.** Under review at *ICLR 2027*. arXiv:2609.32942.
- Baser, O., Sozen, M., **Kale, K.**, Chinchali, S., & Vishwanath, S. **RoboShape: Information-Theoretic Point Cloud Representations for Privacy-Aware Robot Perception.** Under review at *TMLR*. arXiv:2608.21380.
- Zhong, N., Tanriverdi, A. E., **Kale, K.**, & Vishwanath, S. **VeryTrace: Verifying Reasoning Traces through Compilable Formalism and Structured Verification.** arXiv:2606.24124, 2026.
- Roberts, J., Mylonakis, K., Roy, S., & **Kale, K.** **Learning Obfuscations of LLM Embedding Sequences: Stained Glass Transform.** arXiv:2506.09452, 2025.
- **Kale, K.**, Mylonakis, K., Roberts, J., & Roy, S. **BeamClean: Language Aware Embedding Reconstruction.** arXiv:2505.13758, 2025.

Skills

ML Systems: PyTorch (advanced), distributed training (FSDP, expert parallelism), SLURM, vLLM
Programming: Python, C++, Hugging Face, Linux
Research: Mechanistic interpretability, representation learning, information theory

Honors & Awards

- Tekfen Holding Merit & Need-Based Scholarship, for academic performance 2020 – 2025
- Ranked **301st** among 2.2 million students (top 0.02%) in the Turkish university entrance exam 2020
- **3rd** place nationally, Fizy National Music Contest 2018

Additional

Languages: Turkish (Native), English (Fluent; TOEFL iBT 112/120), Italian (Intermediate)
Interests: Piano (9 years of conservatory training), music production, tennis, kiteboarding